

PomProc Relational Database Architecture & Data Guide

This document explains in simple, clear language how the PomProc database tables work, how they link together, and step-by-step rules for filling them out correctly in Excel or Odoo.

1. What is a Relational Database? (Simple Concept)

Think of a relational database like a set of interconnected spreadsheet sheets where every row has a unique passport photo, called a **Primary Key (PK)**. When one sheet wants to reference an item from another sheet, it uses that item's passport number, called a **Foreign Key (FK)**.

- **Primary Key (PK):** A unique code that identifies one specific row (e.g., Product SKU MAT-MOP-40 or Category Outillage).
- **Foreign Key (FK):** A reference stored in Table B that points directly to the Primary Key in Table A.
- **Relationship (1 to Many):** One Category can have many Products, but each Product belongs to only one Category.
- **Relationship (Many to Many):** One Kit contains many Products, and one Product can be inside many Kits.

2. How the Tables Connect Together

Below is a map showing how data flows across the 6 core tables in the PomProc system:

Source Table	Points To (Target Table)	Link Field (Foreign Key)	Why They Connect
Products	Categories	catId / Name	Assigns colors, icons, asset number rules, and size/color variants to products.
Kits	Products	ItemSKU	Pulls real product details (price, packaging, stock entity) into standard work bundles.
Templates	Products & Users	ItemSKU & Owner	Links user-saved order templates to specific products and creator accounts.
Orders	Users	username	Connects site requests to the person who created the order and the team leader.
Order Lines	Orders & Products	order_id & ItemSKU	Lists individual product items requested on a specific warehouse order.

3. Detailed Breakdown of Every Table

Table 1: Categories (product_categories)

Purpose: Group products logically on screen (e.g., EPI, Outillage, Machines) and control special options like whether serial numbers are required.

Column Name	Description	How to Fill It Out
Name (PK)	Unique name of the category.	Write clean text: EPI, Machines, Consommables.
DisplayOrder	Sort order in the mobile app.	Numbers starting from 1 upwards (1, 2, 3...).
RequiresAssetNum	Forces warehouse staff to scan/type an asset #.	Enter yes for machinery/tools; no for consumables.
Color & Icon	Visual badge styling in the interface.	Hex color code (e.g., #E8792B) and icon name (e.g., mac, epi).
Variants	Option drop-downs (sizes, colors).	Format: Taille: S, M, L Couleur: Blanc, Noir. Leave empty if none.

Table 2: Products (products)

Purpose: Master inventory table holding every item that can be ordered or tracked.

Column Name	Description	How to Fill It Out
SKU (PK)	Strictly unique internal product ID.	Use structured code: MAC-KAR-500 or 710522 (must never repeat!).
Reference	Supplier catalog number.	Supplier code. Can repeat if multiple size options share one supplier ref.
Nom	Full readable item title.	Descriptive text: Mop Uniko Microtex - 40 cm.
CatId (FK)	Links to Category Name.	Must match an exact Name from the Categories table (e.g., Matériel).
Unite & QteEmballage	Unit of measure and pack size.	Unite: Pièce, Bidon, Paquet. QteEmballage: number (e.g., 25).
Prix	Unit purchase price in Euros.	Decimal number (e.g., 7.71). Use 0.0 if unpriced.
Entity	Default billing company.	Choose one: IIS (internal), FFA, or DC.
Assoc	"À prévoir aussi" linked products.	Comma-separated SKUs of mandatory accessories (e.g., Karcher requires hose SKU).

Table 3: Kits (kits)

Purpose: Pre-packaged equipment bundles (e.g., "Kit nettoyage vitres") that let users add multiple items to a request in one click.

Column Name	Description	How to Fill It Out
KitName	Title of the pre-made kit.	Repeated for every product inside the kit (e.g., Kit démarrage chantier).
ItemSKU (FK)	Product included in the kit.	Must match a valid SKU in the Products table.
Variant	Pre-selected size/color.	Specific option (e.g., M / Bleu). Leave blank if product has no variant.
Qty & Mode	Quantity and packaging type.	Qty: number (e.g., 2). Mode: unite or paquet.
Entity (FK)	Billing entity override for this line.	Set to FFA, IIS, or DC. Overrides product default if needed.

Table 4: Templates (templates)

Purpose: Saved routine checklists created by supervisors or admins for standard site types.

Column Name	Description	How to Fill It Out
TemplateName	Name of the saved template.	Group key (e.g., Entretien Bureaux Hebdo).
Owner (FK)	Creator username.	Must match a valid username in the Users table.
Shared	Visibility flag.	yes (available to all users) or no (private to owner).
ItemSKU (FK) & Qty	Item code and count.	Valid product SKU and count (e.g., 4).

Table 5: Users (users)

Purpose: System user accounts, security roles, and contact info for automated notifications.

Column Name	Description	How to Fill It Out
username (PK)	Unique account ID.	Format: prenom.nom or admin.
role	Access level.	user (Demandeur), magasinier (Warehouse), or admin.
workplace	Default site/agency location.	Text location: IIS Stock Central or Chantier Louise.
email & phone	Contact details.	Used for instant WhatsApp/Email dispatch alerts.

Table 6: Orders & Order Lines (orders & order_lines)

Purpose: Live warehouse orders submitted by field personnel, tracked through preparation to final completion and invoicing.

- **Header Data:** Unique Ref (e.g., 2609-001), Site Name, Site Number, Departure Date/Time, Status (envoyee → en_cours → terminee).
- **Line Data:** Items requested vs. items actually prepared by warehouse staff, missing items, replacement items, and asset serial numbers logged.

4. Summary of Data Filling Rules for Excel & Odoo

Golden Rules to Prevent Database Errors:

1. **Format All Codes as Text:** In Excel, always set SKU, Reference, and Site Number columns to **Text** formatting so zeroes like 00142 are not lost.
2. **Check Foreign Key Spelled Matches:** If a product belongs to category Outillage, the category sheet must contain the exact word Outillage. A typo like outillage will cause database import failures.
3. **Maintain SKU Uniqueness:** No two products may share the same SKU. If supplier references repeat across sizes, create composite SKUs like MOP-710522-40CM.